House Price Prediction – Task 1 Summary

Overview

The objective of this project was to develop a machine learning model capable of predicting house prices based on various property characteristics. The analysis was performed using a housing dataset containing **545 records and 13 features**, including area, bedrooms, bathrooms, parking facilities, furnishing status, and other amenities.

Methodology

The dataset was explored, cleaned, and prepared for analysis by checking for missing values, removing duplicate records, and converting categorical variables into numerical form using one-hot encoding. The data was then split into training and testing sets using an 80:20 ratio.

Two regression models were developed and compared:

- Linear Regression
- Random Forest Regressor

Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

Model	MAE	RMSE	R^2 Score
Linear Regression	970,043	1,324,507	0.653
Random Forest Regressor	1,021,546	1,400,566	0.612

Results and Insights

Among the two models, **Linear Regression achieved the best performance**, with an R^2 Score of **0.653**, compared to **0.612** for the Random Forest model. This indicates that the dataset contains relatively strong linear relationships between housing features and property prices.

Feature importance analysis revealed that **area** was the most influential factor affecting house prices, followed by **bathrooms, air conditioning, parking availability, and number of stories**. The results suggest that larger homes with better amenities generally command higher market values.

One interesting observation was the noticeable impact of comfort-related features such as air conditioning and furnishing status, highlighting that buyer preferences extend beyond property size alone.

Conclusion

This project successfully demonstrated the use of machine learning techniques for house price prediction. The findings show that property size and key amenities play a major role in determining market value. Based on the evaluation results, Linear Regression proved to be the most effective model for this dataset and can serve as a useful tool for supporting data-driven pricing decisions in the real estate sector.